SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

$C_{03}-A1_3$ - cloud computing CSA0520

SHORT Answer Test.

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(1) Define cloud computing and explain any four characteristics that Makes it different from traditional on-Premise.

(2) Trace the Evolution of cloud computing - from Hardware Evolution to internet Software Evolution, and Show how these changes Enabled Modern cloud Platforms.

3. Explain the role of Server Virtualization as an cloud computing Differential between Virtualization as an Enabling technology and cloud Computing as a Service Model.

4. Discuss the role of data Centers in delivering cloud Services.

Differentiate IaaS, PaaS, SaaS and Xaas with one Suitable real-world example for Each Model

(1) Definition:-

Cloud computing in a delivered by cloud service Software, storage and Network and internet these is called cloud computing.

Characteristics:- four

- on-demand
- Scalability
- on-as-pre-use
- Internet

- on-demand :-

Cloud computing is gives flexibity of a service data over cloud and use the service only Needed.

→ Tranditional computing requires arranged to has Maintance and Manage the Software, storage Network and internet.

(2) Evolution of Cloud Computing :-

→ cloud computing developed through services in stages.

- Mainframe Evolution :-

→ Early computers were horge and experimen

→ Mainframe allowed Multiples users to share Computing resources.

Network Computing:-

→ Computer Network Connected different System

Virtualization :-

→ Virtualization allowed an physical Services to runs Multiple Virualization.

→ This improved Hardware.

→ Internet Software Evolution.

→ web application, and web-based Services, because, Virtualization

→ Modern cloud Platform

→ Data Centers Combined flight speed Network, Virtualization Connected throught Internet Technology.

(3) Role of Servers Virtualization Cloud Computing:-

→ Servers Virtualization is a technology the divides one physical into Multiple virtualizatim Services and cloud Virtualization.

Role:-

→ It allows Multiple applications to run on one physical Services.

→ It improved the virtualization servers.

Example:-

one physical server can be 4 virtual Machine each working.

Virtualization	cloud computing:-
Technology	Service Model
Creates virtual Machines	Provides computing Services.
uses physical resources efficiently	Provide resources through Internet.
can work without a Public cloud	usually uses virtualization and other technologies.

- Stores uses data and applications
- Provided computing Powers.

(11) Role of data Centers is delivering cloud Servics :-

A data Centers is a facility containing many servers, storage systems, Networking Equipment and security systems.

Role:-

- Connects cloud Services to the internet

- Stores user data and applications.
- Provides computing power.
- Provides backup and recovery.
- Maintains security, and availability.
- Allows cloud Providers to serve many users

Example :-

Cloud Provider operate large the data centers. Provides servers to users around the world of these.

- A data center is a facility containing many of servers, storage system, Networking, equipment system.

Differentiable IaaS, PaaS, SaaS, and XaaS:-

Model	Meaning	What is Provided	Example
IaaS	Infrastructure as a Service	virtual servers, storage, Networks	AWS EC2
PaaS	Platforms as Service	Platform for developing application	Google APP

SaaS	Software as Service	Ready-to-use software	Gmail
XaaS	Anything / Everything as Service	Almost any IT service through the internet	Cloud storage, AI, as Service

IaaS → Infrastructure

PaaS → Platform

SaaS → Software

XaaS → Anything as Service.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow